Stat107_Final

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R Markdown

Dataset before cleaning and pre processing

```
covid <- read.csv("~/Desktop/Stat107/Stat107Final/STAT107-Project/Provisional_COVID-19_Deaths_by_Sex_an
```

How many observations our data set contains

```
dim(covid)
```

```
## [1] 137700     16
```

The column names in the data set

```
colnames(covid)
```

```
##  [1] "Data.As.Of"
##  [2] "Start.Date"
##  [3] "End.Date"
##  [4] "Group"
##  [5] "Year"
##  [6] "Month"
##  [7] "State"
##  [8] "Sex"
##  [9] "Age.Group"
## [10] "COVID.19.Deaths"
## [11] "Total.Deaths"
## [12] "Pneumonia.Deaths"
## [13] "Pneumonia.and.COVID.19.Deaths"
## [14] "Influenza.Deaths"
## [15] "Pneumonia..Influenza..or.COVID.19.Deaths"
## [16] "Footnote"
```

Overall Summary of the dataset

```
summary(covid)
```

```
##   Data.As.Of         Start.Date          End.Date            Group
##  Length:137700      Length:137700      Length:137700      Length:137700
##  Class :character   Class :character   Class :character   Class :character
##  Mode  :character   Mode  :character   Mode  :character   Mode  :character
##
##
##
##
##       Year          Month          State               Sex
##  Min.   :2020   Min.   : 1.0    Length:137700      Length:137700
##  1st Qu.:2020   1st Qu.: 3.0    Class :character   Class :character
##  Median :2021   Median : 6.0    Mode  :character   Mode  :character
##  Mean   :2021   Mean   : 6.2
##  3rd Qu.:2022   3rd Qu.: 9.0
##  Max.   :2023   Max.   :12.0
##  NA's   :2754   NA's   :13770
##   Age.Group         COVID.19.Deaths    Total.Deaths       Pneumonia.Deaths
##  Length:137700      Length:137700      Length:137700      Length:137700
##  Class :character   Class :character   Class :character   Class :character
##  Mode  :character   Mode  :character   Mode  :character   Mode  :character
##
##
##
##
##  Pneumonia.and.COVID.19.Deaths Influenza.Deaths
##  Length:137700                 Length:137700
##  Class :character              Class :character
##  Mode  :character              Mode  :character
##
##
##
##
##  Pneumonia..Influenza..or.COVID.19.Deaths   Footnote
##  Length:137700                            Length:137700
##  Class :character                         Class :character
##  Mode  :character                         Mode  :character
##
##
##
##
```

Cleaning the Data

Anaylzing how many values are missing inside the data set

```
sum(is.na(covid))
```

```
## [1] 16524
```

Replacing the missing values with 0

```r
for(col in 1:ncol(covid)){
  for(row in 1:nrow(covid)){
    if(is.na(covid[row,col]) || is.na(covid[row,col]) == " "){
      covid[row,col] <- 0
    }
  }
}

write.csv(covid, "covid_cleaned.csv", row.names = FALSE)
```

Final Datset after cleaning and pre processing

```r
covid_cleaned <- read.csv("~/Desktop/Stat107/Stat107Final/STAT107-Project/covid_cleaned.csv")
sum(is.na(covid_cleaned))
```

```
## [1] 0
```